

widens as you switch to the older DirectX 11 API. With Vulkan 1.1, the NVIDIA cards perform much better leaving the 9070 XT behind, especially when ray-tracing is involved. In 3DMark Port Royale, which is a ray-tracing benchmark, the RTX 4080 Super and RX 9070 XT seem evenly matched and the 5070 Ti takes a slight lead. On the Linux front, RADV which is the open-source implementation of Vulkan for AMD GPUs, is being optimised and more optimisations will be added in the days to come.

On the gaming front, the RX 9070 XT beats the RTX 4080 Super and 5070 Ti and equals the RX 7900 XTX in Assassin's Creed Mirage. And in F1 24, it nearly matches up to the 7900 XTX. However, in Cyberpunk 2077, Hogwarts Legacy and The Witcher 3, the 5070 Ti either equals or bests the 9070 XT. So you can say that the RX 9070 XT is a nice competitor to the 5070 Ti and the pricing can really make or break the card. Ideally, it should be priced lesser than the RTX 5070 Ti since NVIDIA uses GDDR7 and has a lead in ray-tracing performance.

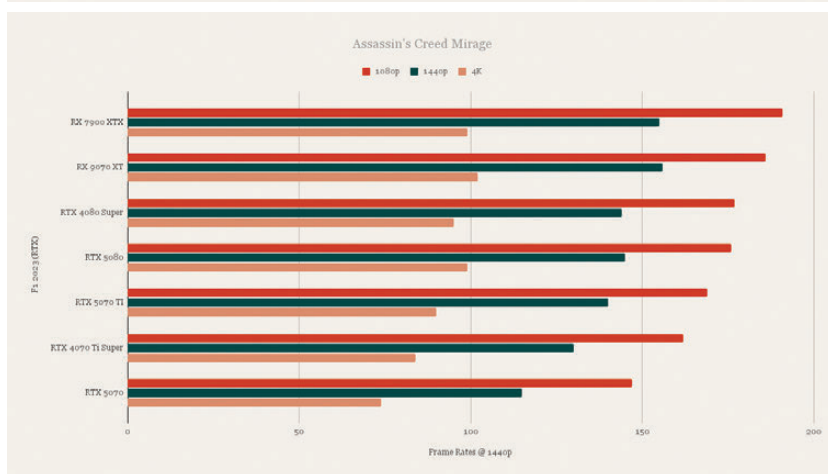
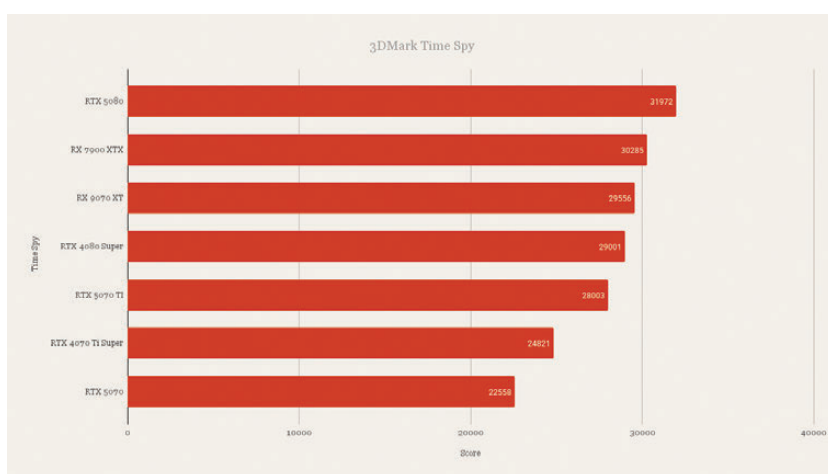
POWER CONSUMPTION AND THERMALS

So what's the secret behind the improved performance of the AMD Radeon RX 9070 XT? Ridiculous power consumption! Peak power draw on the Radeon RX 9070 XT can hit about 400 Watts during gaming and even more so if you push all the paddles, which in comparison to the 300 Watts of the RTX 5070 Ti and the 250 Watts on the RTX 5070 is quite inflated.

We found the Sapphire PURE RX 9070 XT to remain relatively cool during gaming with peak temperatures on the GPU barely hitting 60 degrees Celsius. However, the memory chips seem to be constantly running at 90 degrees Celsius which is quite high.

OVERCLOCKING

With clock speeds touching 3 GHz and power consumption also crossing into the 400 Watt territory, there isn't much headroom to push the card. We got 103



MHz above the default clock speeds but the synthetic performance dropped by a meagre amount. For the unit we received, it was better off to leave the card at the factory overclock settings rather than mess around with the clock speeds.

VERDICT

The Sapphire AMD Radeon RX 9070 XT graphics card is a powerful graphics card that offers excellent performance for 4K gaming and content creation. Based on AMD's latest RDNA 4 architecture, the 9070 XT enjoys significant improvements in efficiency and performance over the previous generation RDNA 3 architecture. However, the compute efficiency comes at a tradeoff with power consumption and that's quite high on the 9070 XT. Nevertheless, with 16 GB of VRAM on a 256-bit interface, improved ray-tracing performance and raster performance that matches and

even beats the 7900 XTX, the Radeon 9070 XT has become a force to be reckoned with. If priced right at around INR 60,000 to 65,000, it offers excellent 4K gaming performance for the money. So if you're looking for an upper-mid-range graphics card that won't break the bank, the RX 9070 XT is a great option but if this card is priced anywhere above 80-85K then the NVIDIA RTX 5070 Ti becomes the better option considering that the performance-per-watt metric shifts in NVIDIA's favour.

—Mithun Mohandas

SPECIFICATIONS

Shader Cores: 4096 | RT Cores: 64 | Tensor Cores: 128 | Base clock: 2400 MHz | Boost clock: 2970 MHz | VRAM: 16 GB | Memory interface: 256-bit | Process node: TSMC N5 / N6 | TDP: 304 W | Architecture: RDNA 4

CONTACT

Company: AMD | Phone: NA | Email: Web form | Website: [nvidia.com/en-in/](https://www.nvidia.com/en-in/)